

Chat Summary – 2025-09-16

This summary captures the key points of our discussion so that you can continue the work seamlessly in a new ChatGPT session.

Spatial Sampling and Distance Optimization

- We reviewed **Moran's I**: a statistic for detecting spatial autocorrelation.
 - Values near +1 → clustering of similar values.
 - Values near -1 → checkerboard (dissimilar adjacency).
 - Values near 0 → random spatial distribution.
 - Used for residual diagnostics and exploratory spatial analysis.
 - We discussed **Response Surface Sampling (Lesch, 2005)**:
 - Originally optimizes over **all pairwise distances** to maximize spatial spread.
 - This is computationally burdensome ($O(n^2)$ memory).
 - Explored using **nearest-neighbor (NN) methods** with KDTree as a substitute:
 - **Pros**: Faster, memory-efficient ($O(n \log n)$).
 - **Cons**: Approximates global optimization; may miss distant pairwise spacing.
 - **Valid if assumptions hold** (uniform data, short correlation range, residuals locally structured).
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Assumptions for NN Approximation

NN-based optimization is effective if:

1. Data are **uniformly or systematically spaced** (drone pixels, EM38 transects).
2. **Short spatial correlation range** dominates (e.g., 30–60 m for soils/vegetation).
3. Residuals are **locally correlated**, not global.
4. PCA space is **convex and isotropic** (sensor correlations stabilized).

These are often true in your datasets.

Methods to Estimate Optimal NN Distance

1. Empirical Sweep

- Test multiple NN thresholds (20–100 m).
- Evaluate R^2 , RMSE, Moran's I, geoMSD.
- Select the elbow point.

2. Variogram of Residuals

- Fit variogram.

- Use ~95% of effective range as NN distance.
- Grounded in geostatistical theory.

3. **Heuristic Local Variance**

- Use KDTree queries to compute local variance across k-NNs.
- Pick distance where variance stabilizes.
- Very RAM-friendly.

4. **Hybrid Validation**

- Apply NN method at chosen distance.
- Compare against full SRS: geoMSD, Moran’s I, prediction accuracy.
- If close, adopt NN for scalability.

 Summary Table

Method	Estimate of Optimal d_{nn}	RAM Friendly	Rigor	Notes
Empirical sweep	Based on R^2 , Moran’s I	 Moderate	☒ Empirical	Best tradeoff if you can afford small loops
Variogram (residuals)	Use 95% of estimated range	 Medium	☒ Theoretical	Matches Lesch (2005) approach
Heuristic (local var)	Distance where variance flattens	☒ Yes	 Approximate	Great for massive datasets with no modeling

 Next Steps for Implementation

- Add a function to your notebook that computes **optimal NN distance** using one or more of the approaches above.
- Validate NN approach against full SRS on a subset of data.
- Document results (geoMSD, Moran’s I, predictive metrics) to demonstrate equivalence.
- If successful, **adopt NN permanently** for memory and runtime efficiency.